Stony Brook Institute at Anhui University

安徽大学纽约石溪学院

Chapter 2 Random variable and their probability distributions

Section 2.4 2-D random variable and its distribution

- 2-D random variable and its distribution
- 2-D discrete random variables
- 2-D continuous random variables
- Common distributions

Section 2.4 2-D random variable and its distribution

Introduction

There are many random experiments that involve more than one r.v.. For example, an educator may study the joint behavior of grades and time devoted to study; a physician may study the joint behavior of blood pressure and weight. Similarly an economist may study the joint behavior of business volume and profit. In fact, most real problems we come across will have more than one underlying r.v. of interest.

Section 2.4 2-D random variable and its distribution

Jointly distributed random variables

Definition Let E be an experiment, and sample space is S. X and Y are two random variables on sample space S.
We call $(X; Y)$ **two-dimensional r.v.** or **two-dimensional random vector**.

For convenience, we call a single r.v. X one-dimensional r.v..
We are going to restrict ourselves to the case where both X and Y are either discrete or of the continuous type. The concepts of probability distribution, distribution function, and PDF are defined by a straightforward generalization of the definition of these concepts in the previous chapter.

2-D random variable and its distribution

Examples

(1)The position of the cannonball's impact point e, i.e. $(\boldsymbol{X},\boldsymbol{Y})$ is a two-dimensional random variable.

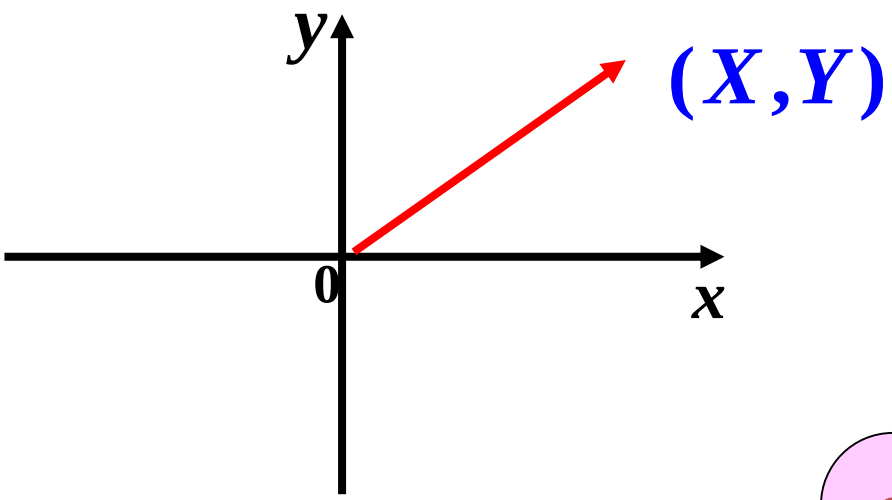
(2)When the development of preschool children in a certain area is examined, the height $\boldsymbol{H}$ and weight $\boldsymbol{W}$ of the children constitute two-dimensional random variables $(\boldsymbol{H},\boldsymbol{W})$.

Note:

The properties of 2-dimensional random variables $(\boldsymbol{X}, \boldsymbol{Y})$ are **not only** related to $\boldsymbol{X}$ and $\boldsymbol{Y}$, **but also** depend on the relationship between these two random variables.

Section 2.4 2-D random variable and its distribution

▲ Geometric meaning of (X,Y)



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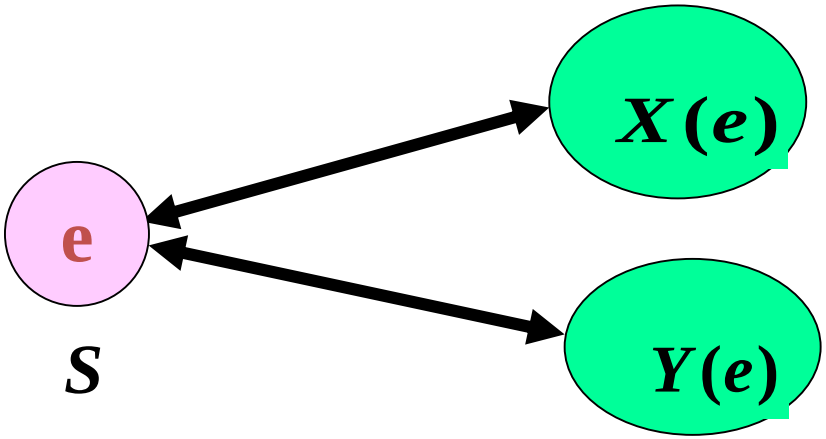


(X,Y) ⟺ A random point (random vector) on a plane

2-D random variable and its distribution

The joint CDF of 2-d random variable

(Joint cdf) Let (X, Y) be two-dimensional r.v., for any real numbers x, y, we define

$$F(x, y) = P\{(X \leq x) \cap (Y \leq y)\} = P\{X \leq x, Y \leq y\}$$

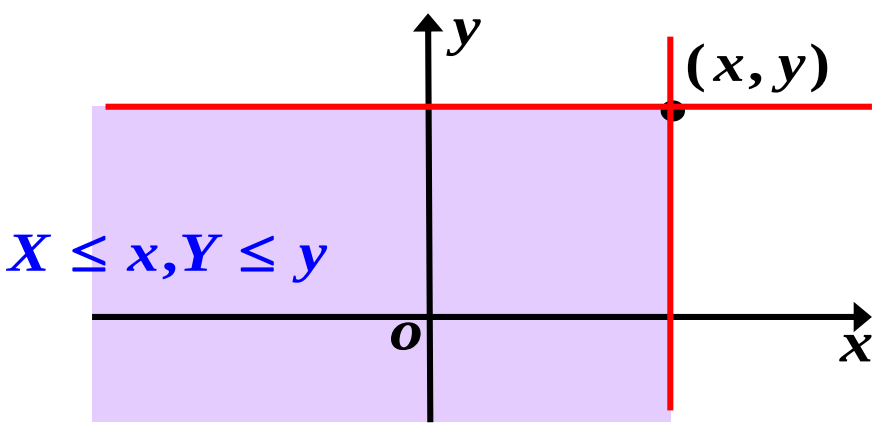
Two-component function $F(x, y)$ is called the joint cumulative distribution function of two-dimensional r.v. (X, Y). In fact, the function $F(x, y)$ shows a probability that random points (X, Y) fall in a closed region $\{X \leq x, Y \leq y\} \subset \mathbb{R}^2$.

What does $F(x, y)$ mean?

2-D random variable and its distribution

The joint CDF of 2-d random variable

What does $F(x, y)$ mean?



The probability of a random trial falls in the purple area.

2-D random variable and its distribution

Properties of $F(x,y)$:

(1) $F(x, y)$ is a non-decreasing function about every variable x and y, i.e.
$F(x, y_1) \leq F(x, y_2)$ if $y_1 < y_2$ for any fixed x;
$F(x_1, y) \leq F(x_2, y)$ if $x_1 < x_2$ for any fixed y.

(2) $0 \leq F(x, y) \leq 1$ and

$$F(-\infty, y) = \lim_{x \to -\infty} F(x, y) = 0$$

$$F(x, -\infty) = \lim_{y \to -\infty} F(x, y) = 0$$

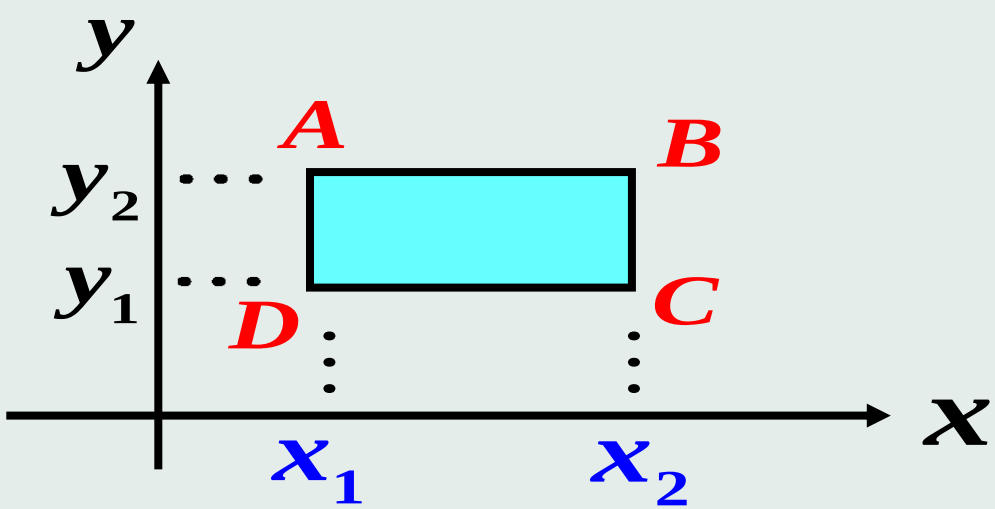
$$F(+\infty, +\infty) = \lim_{x, y \to +\infty} F(x, y) = 1$$

2-D random variable and its distribution

Properties of $F(x,y)$:

(3) For any $x_1 < x_2$ and $y_1 \leq y_2$, the probability that (X, Y) belongs to a given rectangle is

$$P(x_1 \leq X \leq x_2, y_1 \leq Y \leq y_2) =$$



2-D random variable and its distribution

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$$F(x_2, y_2) - F(x_2, y_1) - F(x_1, y_2) + F(x_1, y_1) \geq 0$$

2-D random variable and its distribution

Properties of $F(x,y)$:

(4) $F(x, y)$ is right continuous function about every variable x and y, i.e.

$$F(x+0, y) = F(x, y), F(x, y+0) = F(x, y)$$

2-D random variable and its distribution

- Properties (1-2-4) are the same as the properties of one-dimensional random variable distribution function.
- Property (3) is different from the distribution function of one-dimensional random variables.
- **If property (1-2-4) are satisfied, but property （3）is not satisfied, it can not be called joint distribution function.**

2-D random variable and its distribution

For example:

$$F(x,y)=\begin{cases}1 & x+y\geq 0\\ 0 & x+y<0\end{cases}$$

Judge whether $F(x,y)$ may be the joint cdf of a random vector.

Solution :

$$(x_2,y_2)=(1,\ 1);\qquad (x_1,y_2)=(-1,1)$$

$$(x_2,y_1)=(1,-1);\qquad (x_1,y_1)=(-1,-1)$$

$$F(1,1)-F(-1,1)-F(1,-1)+F(-1,-1)$$

$$=1-1-1+0=-1<0$$

This shows that $F(x,y)$ is **not** a joint distribution function of two-dimensional random variables, but only a binary function.

Section 2.4 2-D random variable and its distribution

- 2-D random variable and its distribution
- 2-D discrete random variables
- 2-D continuous random variables
- Common distributions

2-D discrete random variable

Definition 2.4.3: If the possible values of a two-dimensional random variable (X, Y) are finite or infinite but can be listed, (X, Y) is called a two-dimensional discrete random variable.

The distribution of 2-D discrete random variable:
The possible values of discrete two-dimensional r.v. (X, Y) are (x_i, y_j), I, j = 1, 2, · · · : We call

$$\boldsymbol{P\{X = X_i, Y = Y_j\} = p_{ij}, \qquad i, j = 1, 2, \ldots}$$

the joint probability distribution or joint distribution law of discrete two-dimensional r.v. (X, Y).

2-D discrete random variable

The distribution of 2-D discrete random variable:

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the joint probability distribution or joint distribution law of discrete two-dimensional r.v. (X, Y).

Properties:

(1) $p_{ij} > 0$

(2) $\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} p_{ij} = 1$

$Y \backslash X$	x_1	x_2	$\cdots$	x_i	$\cdots$
y_1	p_{11}	p_{21}	$\cdots$	p_{i1}	$\cdots$
y_2	p_{12}	p_{22}	$\cdots$	p_{i2}	$\cdots$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	
y_j	p_{1j}	p_{2j}	$\cdots$	p_{ij}	$\cdots$
$\vdots$	$\vdots$	$\vdots$		$\vdots$	

2-D discrete random variable

Example 2.4.1: The possible value of $r.v.$ X is 1, 2, 3 and 4, and the four possible values are equally likely. Y is a $r.v.$ and its possible values are integers among [1, X]. Calculate the probability distribution of (X, Y).

Solution:

Y \ X	1	2	3	4
1	$\frac{1}{4}$	$\frac{1}{8}$	$\frac{1}{12}$	$\frac{1}{16}$
2	0	$\frac{1}{8}$	$\frac{1}{12}$	$\frac{1}{16}$
3	0	0	$\frac{1}{12}$	$\frac{1}{16}$
4	0	0	0	$\frac{1}{16}$

$$P\{X=i,Y=j\}=P\{Y=j|X=i\}P\{X=i\}=\frac{1}{i}\cdot\frac{1}{4},\ i=1,2,3,4,\ j\le i.$$

2-D discrete random variable

Example 2.4.2: There are 3 blue, 2 red and 3 green pens in a black box, pick two of the pens randomly. Let X and Y be the number of picked blue and red pens. Calculate the probability distribution of (X, Y)

Solution:

The possible value of (X,Y) are:

(0,0), (0,1),(0,2),(1,0),(1,1),(2,0).

2-D discrete random variable

Example 2.4.2: There are 3 blue, 2 red and 3 green pens in a black box, pick two of the pens randomly. Let X and Y be the number of picked blue and red pens. Calculate the probability distribution of (X, Y)

Solution:

$$P\{X=0,Y=0\}=\binom{3}{0}\cdot\binom{2}{0}\cdot\binom{3}{2}\Big/\binom{8}{2}=\frac{3}{28},\qquad P\{X=1,Y=0\}=\binom{3}{1}\binom{2}{0}\binom{3}{1}\Big/\binom{8}{2}=\frac{9}{28},$$

$$P\{X=0,Y=1\}=\binom{3}{0}\binom{2}{1}\binom{3}{1}\Big/\binom{8}{2}=\frac{3}{14},\qquad P\{X=1,Y=1\}=\binom{3}{1}\binom{2}{1}\binom{3}{0}\Big/\binom{8}{2}=\frac{3}{14},$$

$$P\{X=0,Y=2\}=\binom{3}{0}\binom{2}{2}\binom{3}{0}\Big/\binom{8}{2}=\frac{1}{28},\qquad P\{X=2,Y=0\}=\binom{3}{2}\binom{2}{0}\binom{3}{0}\Big/\binom{8}{2}=\frac{3}{28}.$$

2-D discrete random variable

Example 2.4.2: There are 3 blue, 2 red and 3 green pens in a black box, pick two of the pens randomly. Let X and Y be the number of picked blue and red pens. Calculate the probability distribution of (X, Y)

Solution:

(0,0), (0,1), (0,2), (1,0), (1,1), (2,0).

Y \ X	0	1	2
0	3/28	9/28	3/28
1	3/14	3/14	0
2	1/28	0	0

Section 2.4 2-D random variable and its distribution

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Continuous two-dimensional r.v.

Definition Let (X, Y) be a two-dimensional r.v., and its joint cumulative distribution function is $F(x, y)$. If there exit a non-negative and integrable function $f(x, y), (x, y) \in \mathbb{R}^2$, satisfy

$$F(x, y) = \int\int_{u \leq x, v \leq y} f(u, v) du dv, \forall (x, y) \in \mathbb{R}^2$$

(X, Y) is called continuous two-dimensional r.v., and $f(x, y)$ is called the joint PDF of (X, Y).

Continuous two-dimensional r.v.

Properties of the joint PDF $f(x, y)$

(1) $f(x, y) \geq 0$;
(2) $\int\int_{\mathrm{R}^2} f(x, y)\mathrm{d}x\mathrm{d}y = 1$
(3) Suppose $G \subset \mathbb{R}^2$, then

$$P\{(X, Y) \in G\} = \int\int_G f(x, y)dxdy$$

(4) If $f(x, y)$ is continuous at point (x, y), then

$$\frac{\partial^2 F(x, y)}{\partial x \partial y} = f(x, y)$$

Continuous two-dimensional r.v.

Example 1

Let (X, Y) be a continuous two-dimensional r.v., and its joint PDF is

$$f(x,y) = \begin{cases} x^2 + kxy, & 0 < x < 1, 0 < y < 2 \\ 0, & \text{elsewhere} \end{cases}$$

(1) Find the constant k;
(2) Find the joint cdf $F(x, y)$;
(3) Find the probability $P\{X + Y < 1\}$.

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Common distributions

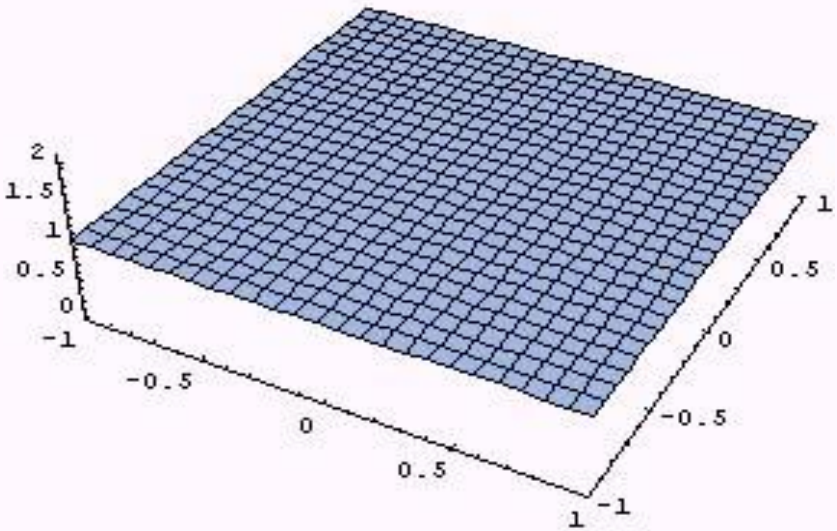
2-D Uniform distribution

Definition 2.4.5: D is a enclosed area in xOy plane with a area of S, if the PDF of 2-D r.v. (X, Y) is

$$f(x,y) = \begin{cases} \frac{1}{S}, (x,y) \in D \\ 0, \ Others \end{cases},$$

Then,

we call (X, Y) has a Uniform distribution in area D.



Common distributions

2-D Exponential distribution

Definition 2.4.5: *if* the PDF of 2-D *r.v.* (X, Y) is

$$f(x,y)=\begin{cases}\lambda e^{-\lambda(x+y)} & x>0, y>0\\ 0 & \text{otherwise}\end{cases}$$

then, 2-D *r.v.*(X, Y) obey the exponential distribution with parameter λ.

Example

Suppose $f(x,y)=\begin{cases} e^{-(x+y)} & x>0, y>0 \\ 0 & \text{otherwise} \end{cases}$

Find : (1) the joint cdf $F(x,y)$;

(2) the probability of (X,Y) in area G. where G surrounded by X axis, Y axis and line $x+y=1$.

Solution

(1) $$F(x,y)=\int_{-\infty}^{x}\int_{-\infty}^{y} f(x,y)dxdy$$

when $x<0, y<0$

$$F(x,y)=\int_{-\infty}^{x}\int_{-\infty}^{y} 0\ dx=0$$

Example

when $x<0, y>0 \quad F(x,y)=0$

when $x>0, y<0 \quad F(x,y)=0$

when $x>0, y>0$

$$F(x,y)=\int_{-\infty}^{x}\int_{-\infty}^{y} e^{-(x+y)}\,dx\,dy$$

$$=\int_{0}^{x} e^{-x}dx\int_{0}^{y} e^{-y}dy=(1-e^{-x})(1-e^{-y})$$

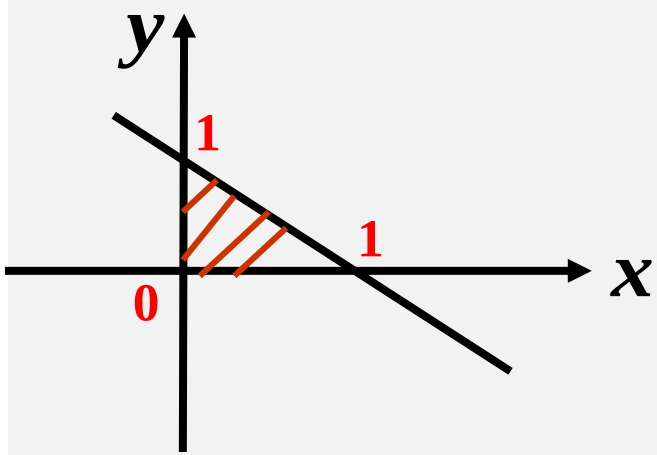
thus,

$$F(x,y)=\begin{cases}(1-e^{-x})(1-e^{-y}) & x>0, y>0\\ 0 & \text{otherwise}\end{cases}$$

Example

(2) Shetch G:

$$f(x,y)=\begin{cases} e^{-(x+y)} & x>0, y>0 \\ 0 & \text{otherwise} \end{cases}$$



$$G: \begin{cases} 0 \le x \le 1 \\ 0 \le y \le 1-x \end{cases}$$

$$P\{(x,y)\in G\} = \iint_G f(x,y)\,dx\,dy$$

$$= \int_0^1 dx \int_0^{1-x} e^{-(x+y)} dy$$

$$= 1 - 2e^{-1} \approx 0.2642$$

Common distributions

2-D Normal distribution

Definition 2.4.5: *If* the PDF of 2-D r.v. (X, Y) is

$$f(x,y)=\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}e^{\frac{-1}{2(1-\rho^2)}\left[\frac{(x-\mu_1)^2}{\sigma_1^2}-\frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2}+\frac{(y-\mu_2)^2}{\sigma_2^2}\right]}$$

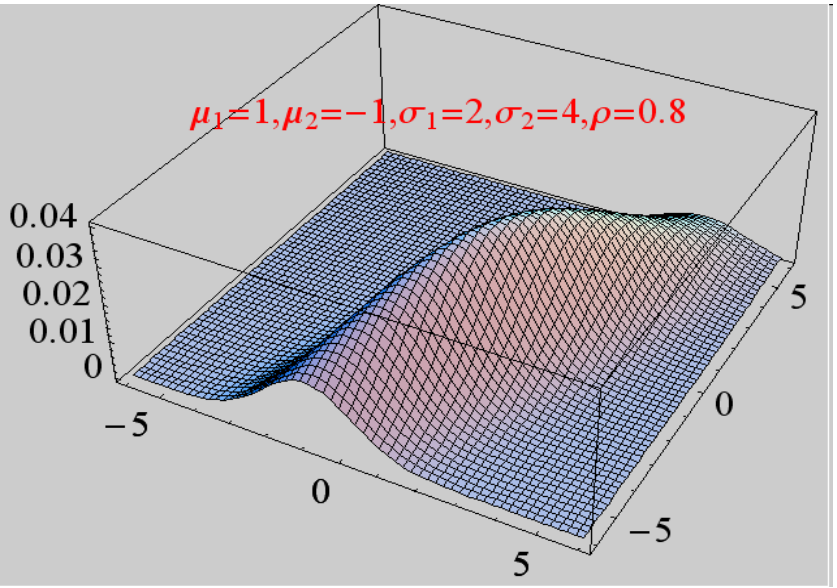
$(-\infty < x < +\infty, -\infty < y < +\infty)$, in which $\mu_1, \mu_2, \sigma_1, \sigma_2, \rho$ are constant, and $\sigma_1 > 0$, $\sigma_2 > 0$, $-1 < \rho < 1$, then we call (X, Y) has a Normal distribution with parameters $\mu_1, \mu_2, \sigma_1, \sigma_2, \rho$, marked by :

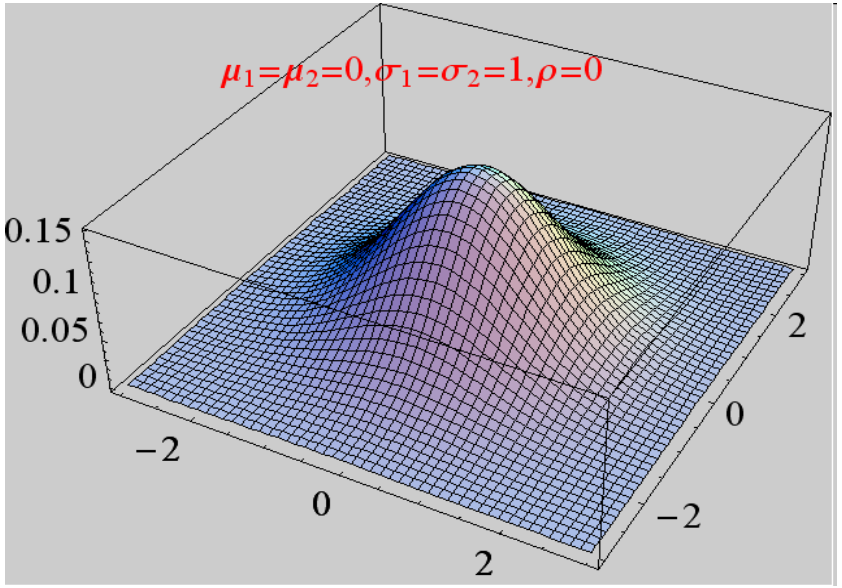
$$(X,Y)\sim N(\mu_1,\mu_2,{\sigma_1}^2,{\sigma_2}^2,\rho)$$

Common distributions

2-D Normal distribution $(X,Y)\sim N(\mu_1,\mu_2,{\sigma_1}^2,{\sigma_2}^2,\rho)$

$$f(x,y)=\frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}}e^{\frac{-1}{2(1-\rho^2)}\left[\frac{(x-\mu_1)^2}{\sigma_1^2}-\frac{2\rho(x-\mu_1)(y-\mu_2)}{\sigma_1\sigma_2}+\frac{(y-\mu_2)^2}{\sigma_2^2}\right]}$$





n-D random variable and its distribution

The above discussion on discrete or continuous random variables can be **extended** to n-dimensional (n > 2) random variables.

n-Dimensional random variable:

Definition 2.4.1: E is a random trial, its sample space is $\Omega = \{e\}$. Assuming that $X_1 = X_1(e)$, $X_2 = X_2(e)$, $X_3 = X_3(e)$,…, $X_n = X_n(e)$, are random variables defined in Ω, then n-dimensional vector $(X_1, X_2, X_3 \ldots, X_n)$ is called a **n-dimensional random vector/variable**.

$$F(x_1, x_2, \ldots, x_n) = P\{(X_1 \le x_1) \cap (X_2 \le x_2) \cap \cdots (X_n \le x_n)\}$$
$$= P\{X_1 \le x_1, X_2 \le x_2, \ldots, X_n \le x_n\}$$

is called a **joint distribution function of random vector** $(X_1, X_2, X_3 \ldots, X_n)$.

New question arise

Note that $(\boldsymbol{X},\boldsymbol{Y})$ is a whole variables, which has a joint-distribution function $\boldsymbol{F}(\boldsymbol{x}, \boldsymbol{y})$.

While $\boldsymbol{X}$ and $\boldsymbol{Y}$ are also random variables, which have distribution functions of $\boldsymbol{F}_{\boldsymbol{X}}(\boldsymbol{x})$ and $\boldsymbol{F}_{\boldsymbol{Y}}(\boldsymbol{y})$, respectively.

So what are their own characteristics?